RAKESH CHOWDARY POLINA

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PROFILE

Software Engineer with 4+ years of experience in designing and optimizing scalable applications using Python and microservices architecture. Expertise in RESTful API development, database optimization, and cloud deployments with AWS and Kubernetes. Proven ability to enhance system performance, streamline CI/CD pipelines, and implement secure, high-availability solutions. Strong background in Agile methodologies and cross-functional collaboration.

EDUCATION

Master of Science in Computer Science **May 2024**
Oklahoma State University, Stillwater, OK **CGPA: 3.909/4**

SKILLS

Languages:	Python, Java, JavaScript, SQL, C, C++, HTML, CSS, Swift
Frameworks:	Flask, Django, React JS, Angular JS, Node JS, Spring boot, FastAPI, React Native
Databases:	MySQL, MongoDB, NoSQL, PostgreSQL, Cassandra
Web Technologies:	RESTful API, Django ORM, SQLAlchemy, OAuth 2, jQuery, JMS, RabbitMQ
Web Services:	REST, SOAP
Tools/Platforms:	Docker, Git, JIRA, Jenkins, Postman, Ansible, Kubernetes, Apache Kafka, Terraform, Swagger
AWS Cloud Services:	EC2, S3, EKS, RDS, ELB, Cloud Watch, CloudFront
Build Tools:	Pip, Poetry

WORK EXPERIENCE

Research Data Analyst, Oklahoma State University **May 2023 – May 2024**

- Engineered and implemented a robust facial recognition pipeline, performing both quantitative and qualitative analysis on image data with Python and InsightFace, to achieve a precision rate of 93.6%
- Implemented DBSCAN clustering and rigorously verified data accuracy across 182,000+ images, enhancing archive reliability and insight precision.
- Designed an intuitive UI for the labeling tool using Flask and React.js, streamlining user engagement and annotations.
- Designed and maintained database structures using SQL, enhancing user interaction management and feedback data integrity, ensuring operational excellence.

Software Engineer, Ericsson, India **Jul 2019 – Jul 2022**

- Transitioned monolithic architecture to microservices using Python, Django, and Flask, designing RESTful APIs for divisional applications. Integrated Apache Kafka for asynchronous communication between services, improving scalability to handle 50% more concurrent users.
- Optimized PostgreSQL and Microsoft SQL Server databases through query tuning, indexing, and Django ORM enhancements, reducing system downtime by 30%. Configured Novell network settings and FTP protocols to ensure secure file sharing across divisional servers.
- Designed and optimized database schemas for divisional applications using Django ORM and SQLAlchemy, enabling seamless data migration and reducing storage costs by 15% through normalization and indexing strategies.
- Designed scalable APIs adhering to REST standards, reducing response time by 40% using Django's asynchronous views and event-driven architecture. Integrated Oracle databases for legacy systems and implemented real-time data synchronization.
- Containerized applications using Docker and orchestrated deployments on VMware vCenter Infrastructure. Leveraged Kubernetes and AWS ECS (public IaaS) to reduce deployment time by 50% and ensure high availability.
- Streamlined CI/CD pipelines with Jenkins and GitLab, utilizing Git for version control. Automated deployments to Apache and Nginx servers, reducing time-to-market by 40%.
- Enforced TCP/IP hardening, AWS IAM roles, and data encryption to meet security standards. Collaborated with cross-functional teams using AGILE (SCRUM) methodologies, including bi-weekly sprints and daily standups.
- Monitored performance of application/database servers (Oracle, SQL Server) using Grafana, resolving bottlenecks in resource allocation. Documented system architectures and APIs to align with divisional information systems requirements.

PROJECTS

Sleep data analysis - Applied LSTM network and Granger causality techniques to investigate sleep data on a dataset comprising over 5,000 individuals' sleep patterns, unveiling insights into temporal dynamics and causal connections within the dataset

HireOSUgrads App - Developed an iOS application for university students and recruiters, featuring intuitive swipe gestures for seamless interactions. Utilized Swift, followed MVC architecture and integrated SQLite for robust data management

Journal App - Developed an android application, that allows users to store their daily events and memories using React Native, Express JS, Redux, Redis, MongoDB, REST APIs, Git.